

Acassist — The Idea in One Page

Offline-first learning and assessment for classrooms the internet leaves behind.

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The problem

In a lot of classrooms, the internet is unreliable exactly when it matters — during a test. Online quiz tools just stop working when the connection drops, so students in under-connected schools can't take assessments the rest of the world takes for granted. That is not a motivation problem. It is an access problem: limited connectivity, weak infrastructure, and a gap between what educators can deliver and what learners can actually reach.

What Acassist is

A desktop app (built with Tauri) where a teacher pushes lessons and quizzes. Students download them once, then use them **fully offline** — lessons are readable and quizzes are taken and **scored right on the device, with no server and no connection needed at test time**.

The novel core — offline, on-device scoring

Every quiz answer ships as a per-question salted hash, never as plaintext in the module file. When a student submits, the native Rust core verifies their answer against that hash locally and returns the score instantly — no network, no answer key sitting readable on disk.

This is honest engineering, not an overclaim: it is **tamper-resistant against casual answer-sharing and cross-device replay** — the real classroom threats — **not "unbreakable" cryptographic security**. Any offline-verifiable design must ship recoverable answers to the device, so a determined student inspecting their own device can't be stopped in *any* offline system; we made the pragmatic tradeoff for the classroom case rather than pretending otherwise.

Offline-first, not anti-online

AI (Groq) is available whenever there's a connection and gracefully degrades when there isn't — it never becomes something that *must* run offline on low-end hardware. Teachers author with AI online; a study aid like "turn a module into flashcards" is AI-enriched online and falls back to a plain render of the existing content offline. **Scoring, though, is always local and deterministic — AI never touches grading.**

Two-sided by design

| Side | Connectivity | What runs |

|-----|-----|-----|

| **Teacher (authoring)** | Online | AI content synthesis (Groq) + guided, non-techy authoring |

| **Student (assessment)** | Offline-first | Take lesson/quiz; device-local scoring. Study aids use AI when online, deterministic offline. Scoring is always local. |

Scoring is *always* local — online is just the easy case. Connectivity never decides whether assessment works.

Built for non-techy teachers (down to elementary)

Teachers are not developers, and some target grade levels reach down to elementary. Authoring is guided, not raw: structured templates and rule-based assists steer the AI, so a teacher never faces a blank prompt box. Content generation runs on **Groq**, online, on the teacher side.

Where modules come from

Teachers build modules aligned to the **DepEd curriculum (MELCs — the government's competency standards)**. The teacher supplies or authors the content they're authorized to teach; Acassist + Groq turn it into offline modules. Acassist is the **tool, not a content library** — it doesn't redistribute DepEd's copyrighted learning materials. (The curriculum competencies themselves are government works and free to align to; the published learning materials are not, so we don't republish them.)

Why it wins

1. **Removes a real barrier** — offline lessons *and* offline scored quizzes for low-connectivity classrooms.
2. **A genuinely novel core** — device-local, offline, tamper-resistant scoring with no server. Not a category clone.
3. **Pragmatic, honest engineering** — a threat-modeled security tradeoff, not a marketing claim.

Likely Q&A

"What about DepEd's copyright?" — We align to the curriculum competencies (MELCs), which are government works. We don't republish DepEd's copyrighted learning materials — teachers bring content they're authorized to use. Acassist is the authoring and delivery tool, not the content.

Roadmap (not built at the event)

Real teacher→student network distribution, accounts and roles, richer lesson authoring and media, online AI study-aid enhancement, and hardened tamper-resistance. The offline-scoring core is the built thing; the platform around it is the roadmap.